



STACK AI × SARVAM AI

TWO ENGINES.

*What Stack can do, what it costs and what it can earn
once both are switched on.*

01

Where things
stand

02

Who does what

03

What Stack gains

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What it costs

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What it can earn

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Build order & risks



THE SHORT ANSWER

Stack AI is written but not switched on. Sarvam AI is not in the code yet. With both live, Stack can summarise, explain and listen in the reader's own language, and the everyday AI jobs cost about a third of what they cost on Claude Haiku.

PART	STATUS	WHAT I FOUND IN THE PROJECT
Stack AI in the app	Built	Four features: Summarise an article, Ask this PDF, Ask your Stack, Tidy a note
Engines in the code	Built	OpenAI and Claude, chosen with one server setting (AI_PROVIDER)
Daily limit per person	Built	50 a day by default, counted in the database. Resets at 00:00 UTC, which is 5:30 AM IST
The ai function on Supabase	Not deployed	Its address answers "function not found" (404)
AI address in the app	Empty	NEXT_PUBLIC_AI_URL is blank, so every AI button is hidden in today's APK
Sarvam engine	Not built	Needs about 40 new lines in one file: supabase/functions/ai/index.ts
Sign-in (AI needs it)	Partly	Email sign-in is on. Google is turned on in Supabase but not yet tested on a phone

THREE DECISIONS

ENGINES USE BOTH	Sarvam for the everyday text jobs, Claude or OpenAI only for PDF chat. Sarvam's chat model reads text, not PDF files, and Stack sends the whole PDF today. Three of the four features move to Sarvam without changing the app (page 3).
COST ABOUT 1/3	A summary costs about ₹0.09 on Sarvam, ₹0.31 on Claude Haiku and ₹1.23 on Claude Opus, which is the default in the code today. Changing the default model before launch matters more than anything else on the cost side (page 5).
PRODUCT 8 NEW THINGS	Sarvam is more than a cheaper chatbot. It also turns speech into text, reads text aloud in Indian voices, translates, and reads scanned pages. That opens eight features that English-only reading apps do not have (page 4).

WHAT THIS GUIDE CANNOT TELL YOU

Stack has no real users yet, so every rupee figure on earnings is a scenario, not a forecast. The cost figures are my estimates from typical text sizes and the listed prices; they were not measured on Stack.



WHO DOES WHAT

Stack AI already has four jobs. Each one should go to the engine that does it well for the least money.

FEATURE IN STACK	WHAT IT DOES	BEST ENGINE	WHY
Summarise an article	Three short points and one "why it matters" line	Sarvam 105B	Text only and cheap. It answers in the article's language
Tidy a note	Fixes spelling, adds a title and structure	Sarvam 105B	Keeps Hindi and Hinglish notes in the writer's own words
Ask your Stack	Answers from up to 30 of your notes, with links to them	Sarvam 105B	Text only. Note links can use the same [1] markers the OpenAI engine uses
Ask this PDF	Answers with links to pages	Claude or OpenAI	Stack sends the PDF file itself (up to about 22 MB). Sarvam's chat model takes text only

WHAT SARVAM SELLS

MODELS AND PRICES FROM SARVAM.AI/API-PRICING AND DOCS.SARVAM.AI, READ ON 01.10.2026

MODEL	JOB	LANGUAGES	LISTED PRICE
Sarvam-105B	Chat: summarise, explain, answer	10 Indian + English	₹29.28 in and ₹73.20 out, per 1 million tokens
Saaras v4	Speech to text	22 Indian + English	₹30 per hour of audio
Bulbul v3	Text to speech	10 Indian + English	₹3 per 1,000 characters
Mayura · Sarvam-Translate	Translation	11 · 23	About ₹4 to ₹5 per 1,000 characters (tiered)
Sarvam Vision	Reads scanned pages into text	22 Indian + English	₹0.50 per page

The 10 Indian languages of the chat model: Hindi, Bengali, Gujarati, Kannada, Malayalam, Marathi, Odia, Punjabi, Tamil and Telugu. Confirm every price in your own Sarvam dashboard before you set limits.

FREE TO TEST

₹100 credit

Every new Sarvam account gets ₹100 of credit. It never expires and all your test phones share it. That is roughly 1,000 summaries.

NO SURPRISE BILL

Prepaid

The Starter plan is pay-as-you-go. When the credit ends, AI stops until you top up. The rest of Stack keeps working.

SPEED LIMIT

40 a minute

Starter allows 40 chat requests a minute on Sarvam-105B. That is enough for a soft launch, tight for a big launch day.



WHAT STACK GAINS

Eight features that become possible once Sarvam is in. None of them exists in the app today. Costs are per use, at the listed prices.

CHAT · SMALL JOB

1 · Summary in your language

A "Summarise in Hindi" choice on any article, also for the other nine languages. The reader picks the language once in Settings.

About ₹0.10 to ₹0.15 each. Hindi text uses more tokens than English.

CHAT · SMALL JOB

2 · Explain this line

Select a hard sentence and tap "Explain simply". Stack answers in plain Hindi, Hinglish or English, and the answer can be saved as a note.

About ₹0.03 each.

SPEECH TO TEXT · MEDIUM JOB

3 · Voice notes

Speak a note in Hindi, Hinglish or any of 22 languages. Saaras writes it down, then "Tidy" cleans it up.

A 30-second note costs about ₹0.25.

SPEECH TO TEXT · MEDIUM JOB

4 · Ask by voice

Ask your Stack or a PDF by speaking instead of typing. It uses the same microphone button as voice notes.

A 10-second question costs about ₹0.08, plus the answer.

TEXT TO SPEECH · MEDIUM JOB

5 · Listen in an Indian voice

Bulbul reads the **summary** aloud in a natural Indian voice. Keep the phone's free built-in voice for full articles: a 1,500-word article would cost about ₹27 on Bulbul.

A summary of 400 characters costs about ₹1.20.

VISION · LARGE JOB

6 · Scanned PDFs become text

Photographed class notes and old scanned books turn into real text, so they can be highlighted, searched and read aloud.

₹0.50 per page, so ₹10 for 20 pages. Limit: 10 requests a minute.

CHAT · LARGE JOB

7 · PDF chat on Sarvam

The phone pulls the text out of the PDF and sends only the pages that match the question. Then PDF chat no longer needs Claude or OpenAI.

Does not work for scanned PDFs without feature 6.

CHAT · MEDIUM JOB

8 · Quiz from your highlights

The morning review adds three questions made from your own highlights, in the language you read in. This was already planned in the business plan.

About ₹0.05 a day per person if the quiz is saved and reused.

WHAT CHANGES FOR THE READER

Today Stack is an English reading app that happens to have Indian news. With these, a student can save an English article, read the points in Hindi, speak a note in Hinglish and hear it back. **Build 1 and 2 first:** they need no new screens, only a language choice and one new button.



WHAT IT COSTS

The same job on three engines. These are estimates from the listed prices at about ₹88 to the dollar. Hindi text costs more than English on every engine because it uses more tokens.

ONE USE OF	SIZE I ASSUMED	SARVAM 105B	CLAUDE HAIKU 4.5	CLAUDE OPUS 5.5 (DEFAULT TODAY)
Summarise an article	1,500 words in, 3 points out	₹0.09	₹0.31	₹1.23
Tidy a note	A long note in and out	₹0.04	₹0.22	₹0.88
Ask your Stack	About 10 notes and a question	₹0.20	₹0.66	₹2.64
Ask this PDF, first question	A 20-page PDF	not possible yet	₹3.70	₹14.80

Claude prices used: Haiku \$1 in and \$5 out, Opus \$4 in and \$20 out per 1 million tokens, as in the business plan PDF. Follow-up questions on the same PDF cost about a tenth of the first one because Claude keeps the PDF cached for a few minutes.

ONE PAYING USER, ONE MONTH

60 AI ACTIONS: 30 SUMMARIES, 10 TIDY, 10 ASK, 10 PDF QUESTIONS ON 3 PDFS

≈ ₹20

Sarvam + Haiku for PDFs. Text jobs cost ₹5, PDF chat costs ₹15.

Recommended

≈ ₹33

Claude Haiku for everything. Close to the ₹30 in the business plan.

The old plan

≈ ₹130

Claude Opus for everything. More than the ₹99 a Plus user pays.

What the code does today

ONE FREE USER, WORST CASE

IF A FREE USER SPENDS THE FULL DAILY LIMIT ON SUMMARIES EVERY DAY FOR 30 DAYS, ON SARVAM

DAILY LIMIT	3 A DAY	5 A DAY	10 A DAY	20 A DAY	50 A DAY (DEFAULT)
Cost per free user per month	₹8	₹14	₹27	₹54	₹135

WHAT THIS MEANS

PDF chat is three quarters of the AI bill. Keep it for paying users, or give free users one PDF question a day. Most people will not use their full limit, so real costs will be lower than the worst case. Set the limit low first and raise it once you see real use.



WHAT IT CAN EARN

The business plan sets **Stack Plus at ₹99 a month** (₹799 a year, students ₹499 a year). Sarvam changes two things: each paying user leaves more money, and Hindi features give people a reason to pay.

₹99

A Plus user pays this each month

– ₹15

Google Play's 15% fee on subscriptions

– ₹20

AI, with Sarvam for text and Haiku for PDFs

≈ ₹64

Left over, before tax and marketing. It was about ₹50 with Haiku only

THREE SCENARIOS

SCENARIOS, NOT FORECASTS. 3% TO 5% OF ACTIVE USERS PAYING IS THE PLAN'S OWN TARGET

ACTIVE USERS	PAYING AT 3%	PLUS INCOME A MONTH	LEFT AFTER FEE AND AI	PAYING AT 5%	LEFT AFTER FEE AND AI
1,000	30	₹2,970	≈ ₹1,900	50	≈ ₹3,200
3,000	90	₹8,910	≈ ₹5,800	150	≈ ₹9,600
10,000	300	₹29,700	≈ ₹19,200	500	≈ ₹32,000

The plan's goal of ₹1 lakh a month by September 2027 needs about 700 Plus users and one institute. At 3% to 5% paying, 700 Plus users means 14,000 to 23,000 active users.

FREE AND PLUS, WITH AI

FREE

Enough to feel the value

- 3 to 5 AI uses a day, on Sarvam
- Summaries and "Explain this line", in English or Hindi
- The phone's own voice for listening
- No PDF chat, or one PDF question a day

Worst case ₹8 to ₹14 per free user per month.

STACK PLUS · ₹99

Everything that costs money to run

- 30 AI uses a day
- PDF chat, Ask your Stack, quiz from highlights
- Voice notes and the Indian voice for summaries
- Sync across devices; scanned PDFs to text

Costs about ₹20 a month in AI for a typical user.

THE RULE THAT KEEPS THIS SAFE

One Plus user pays for about five free users at a limit of 5 a day, and for only one at a limit of 20. So free AI is a marketing cost: keep the free limit small until people start paying. Stack has no payment system yet. Subscriptions inside the Android app must use Google Play Billing, and that is a separate build.



HOW TO SWITCH IT ON

Five steps, each one useful on its own. Stop after any step and Stack still works. "You" means a dashboard or a key only you can reach; "Code" is work in the repo.

A**Put Stack AI live as it is**

YOU · 20 MINUTES

Deploy the function, give it one key and a low daily limit, put its address in `.env.local`, rebuild the app.

```
# in the project folder
npx supabase login
npx supabase link --project-ref <your project id>
npx supabase functions deploy ai --no-verify-jwt
npx supabase secrets set AI_PROVIDER=openai OPENAI_API_KEY=sk-... OPENAI_MODEL=gpt-5.4-mini AI_DAILY_LIMIT=10
# .env.local, then ./build-apk.sh
NEXT_PUBLIC_AI_URL=https://<your project id>.supabase.co/functions/v1/ai
```

B**Add the Sarvam engine**

CODE · HALF A DAY

A new `runSarvam` next to `runOpenAI` in the function. Sarvam's chat API has the OpenAI shape, at `api.sarvam.ai/v1/chat/completions`. Add a setting per feature so PDF chat can stay on Claude or OpenAI while the other three use Sarvam. You add `SARVAM_API_KEY` to the function's secrets.

C**Language features**

CODE · 1 TO 2 DAYS

"AI language" in Settings, "Summarise in Hindi", and "Explain simply" in the selection toolbar. No new Sarvam product is needed, only new prompts.

D**Voice**

CODE · ABOUT A WEEK

A microphone button for voice notes and spoken questions (Saaras), and "Listen to the summary" in an Indian voice (Bulbul). Needs the microphone permission and a new line in the Play Data safety form.

E**Scanned PDFs and PDF chat on Sarvam**

CODE · 1 TO 2 WEEKS

Sarvam Vision for scanned pages, then text-based PDF chat. Do this last: it is the biggest job and PDF chat already works on Claude and OpenAI.

THINGS TO SETTLE BEFORE STEP B

QUESTION

WHAT TO DO

Which model is the default?

The code falls back to `claude-opus-5-5` and `gpt-5.5`. Set a cheaper model in the secrets (`AI_MODEL`, `OPENAI_MODEL`) before any tester gets the app

What does the app call the engine?

The "ask first" sheet shows `NEXT_PUBLIC_AI_ENGINE`. With two engines it must name both, for example "Sarvam AI and Claude"

Is Hindi good enough?

Run 20 real Hindi and English articles through Sarvam and read every summary yourself before you turn it on for others



RISKS & RULES

RISK	WHAT COULD HAPPEN	WHAT TO DO
The bill grows	A few heavy users or a bug use up the credit in days	Low daily limit, cheap default models, prepaid Sarvam credit, and a monthly spend limit on the Claude or OpenAI account
Credit runs out	AI shows "Stack AI is paused: the AI account is out of credit"	Check the Sarvam and Claude dashboards every Sunday. The rest of the app keeps working
Wrong answers	A summary says something the article did not	Always show the page and note links, which Stack already does. Add a "Report this answer" button: Google Play asks for one in apps that make AI content
Privacy	Text leaves the phone and goes to an AI company	Stack already asks before sending anything and has one switch to turn AI off. Name Sarvam on the privacy page and in the Play Data safety form. Read Sarvam's data terms before launch
Sarvam is busy or down	More than 40 requests a minute, or an outage	Fall back to the second engine for that request. The function already has the pieces to swap engines
Voice costs	Reading whole articles aloud on Bulbul costs about ₹27 each	Use Bulbul only for summaries. Keep the phone's built-in voice for long reading
Reset time	The limit resets at 5:30 AM IST, which confuses Indian users	Count the day in Indian time in the database function (ai_take) so it resets at midnight IST
Big PDFs	PDFs over about 22 MB cannot be sent	The app already says so. Text-based PDF chat (step E) removes the limit for normal PDFs

SOURCES

Sarvam prices (read 01.10.2026)

sarvam.ai/api-pricing

Sarvam models and languages

docs.sarvam.ai/api-reference-docs/getting-started/models

Sarvam free credit, plans and rate limits

docs.sarvam.ai/api-reference-docs/getting-started/pricing

Sarvam chat API

docs.sarvam.ai/api-reference/chat/chat-completions-v1

Stack AI as built

`supabase/functions/ai/index.ts · supabase/schema.sql · src/lib/ai.ts`

Plus pricing, Claude prices, targets

StackForge_Stack_Plan.pdf, pages 6, 8 and 12

Live checks on 01.10.2026

The ai function address, the Supabase sign-in settings and `.env.local`

Not checked: Sarvam's data-handling terms, whether sign-up asks for a card, and the quality of its Hindi summaries on Stack's own articles. Related: Stack_Soft_Launch_Plan.pdf and Stack_India_Identity.pdf in the same folder.